Convex optimization for machine learning – TD5

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1 Exercises

Exercise 1. (Support vector machine)

Consider a training set formed by couples $(\mathbf{x}_i, y_i)$ for $i \in \{1, \dots, n\}$ where $\mathbf{x}_i$ is a feature vector in $\mathcal{X}$ and $y_i \in \{-1, +1\}$ for all i. The (affine) hyperplane $\mathcal{H}_{\mathbf{w},b} := \{\mathbf{x} \in \mathbb{R}^d,\ \langle \mathbf{w}, \mathbf{x}\rangle + b = 0\}$ is called *separating* if

$$\forall i, \quad y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) > 0\,.$$

In the sequel, we assume that a separating hyperplane exists. Among all separating hyperplanes, we seek to find the one which maximizes the minimum distance

$$f(\mathbf{w}, b) = \min_{i=1,\dots,n} d(\mathbf{x}_i, \mathcal{H}_{\mathbf{w},b})\,.$$

1. Show that if $(\mathbf{w}, b)$ defines a separating hyperplane, then[1] $f(\mathbf{w}, b) = c(\mathbf{w}, b)/\|\mathbf{w}\|$ where $c(\mathbf{w}, b) = \min_i y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b)$.

Thus, we are interested in solving the problem

$$\max_{\mathbf{w},b} \frac{c(\mathbf{w}, b)}{\|\mathbf{w}\|} \text{ such that } \forall i,\ y_i(\langle \mathbf{w}, \mathbf{x}_i\rangle + b) \geq 0\,. \tag{1}$$

Let $(\mathbf{w}^*, b^*)$ be a solution and define

$$\mathbf{v}^* = \frac{\mathbf{w}^*}{c(\mathbf{w}^*, b^*)} \text{ and } \ a^* = \frac{b^*}{c(\mathbf{w}^*, b^*)}$$

2. Justify that $(\mathbf{w}^*, b^*)$ and $(\mathbf{v}^*, a^*)$ define the same separating hyperplane.
3. Prove that $(\mathbf{v}^*, a^*)$ solves the optimization problem

$$\max_{\mathbf{v},a} \frac{1}{\|\mathbf{v}\|} \text{ such that } \forall i,\ y_i(\langle \mathbf{v}, \mathbf{x}_i\rangle + a) \geq 1\,.$$

4. Deduce that $(\mathbf{v}^*, a^*)$ solves the optimization problem

$$\min_{\mathbf{v},a} \frac{\|\mathbf{v}\|^2}{2} \text{ such that } \forall i,\ 1 - y_i(\langle \mathbf{v}, \mathbf{x}_i\rangle + a) \leq 0\,. \tag{2}$$

5. Write the Lagrangian $L(\mathbf{v}, a; \phi)$.
6. Write the KKT conditions.

[1] Remember the previous tutorial where we proved (or not) that:

$$\mathrm{d}(\mathbf{z}, \mathcal{H}_{\mathbf{w},b}) = \min_{\mathbf{x} \in \mathcal{H}_{\mathbf{w},b}} \|\mathbf{x} - \mathbf{z}\| = \frac{|\langle \mathbf{w}, \mathbf{z}\rangle + b|}{\|\mathbf{w}\|}\,.$$

7. Let $(\mathbf{v}, a; \phi)$ be a saddle point of the Lagrangian. Show that ϕ_i is non-zero only if $y_i(\langle \mathbf{v}, \mathbf{x}_i \rangle + a) = 1$.

The training points $(\mathbf{x}_i, y_i)$ satisfying the above property are the closest to the hyperplane $\mathcal{H}_{\mathbf{v},a}$. The corresponding $\mathbf{x}_i$'s are often called *support vectors*.

8. If one is given a dual solution $\phi^\star$, how to recover a primal solution $(\mathbf{v}^*, a^*)$ from $\phi^\star$?

Define the $n \times n$ matrices $K = (\langle \mathbf{x}_i, \mathbf{x}_j \rangle)_{i,j=1\ldots n}$, $D = \mathrm{diag}(y_1 \ldots y_n)$ and $\mathbf{1}^T = (1, \ldots, 1)$.

9. Prove that the dual problem reduces to

$$\min_{\substack{\phi \geq 0 \\ \langle \mathbf{y}, \phi \rangle = 0}} \frac{1}{2} \langle KD\phi, D\phi \rangle - \langle \mathbf{1}, \phi \rangle \,.$$

10. Assume that this algorithm has identified a dual solution $\phi^\star$. Write explicitely the classifier as a function of $\phi^\star$.

11. What part of the training data do you need in order to implement the above classifier?